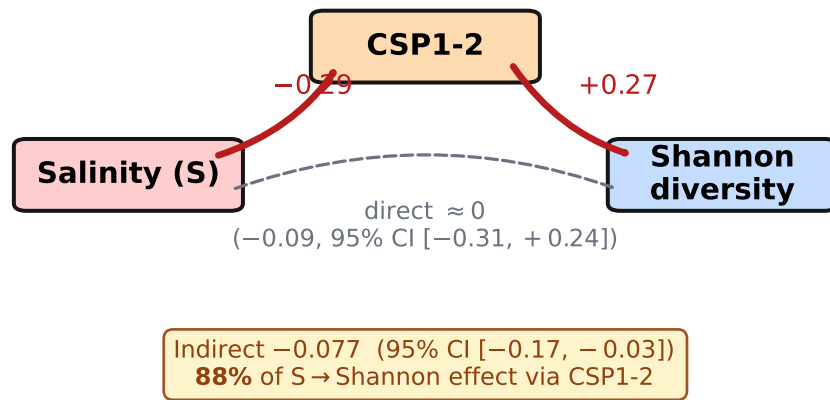
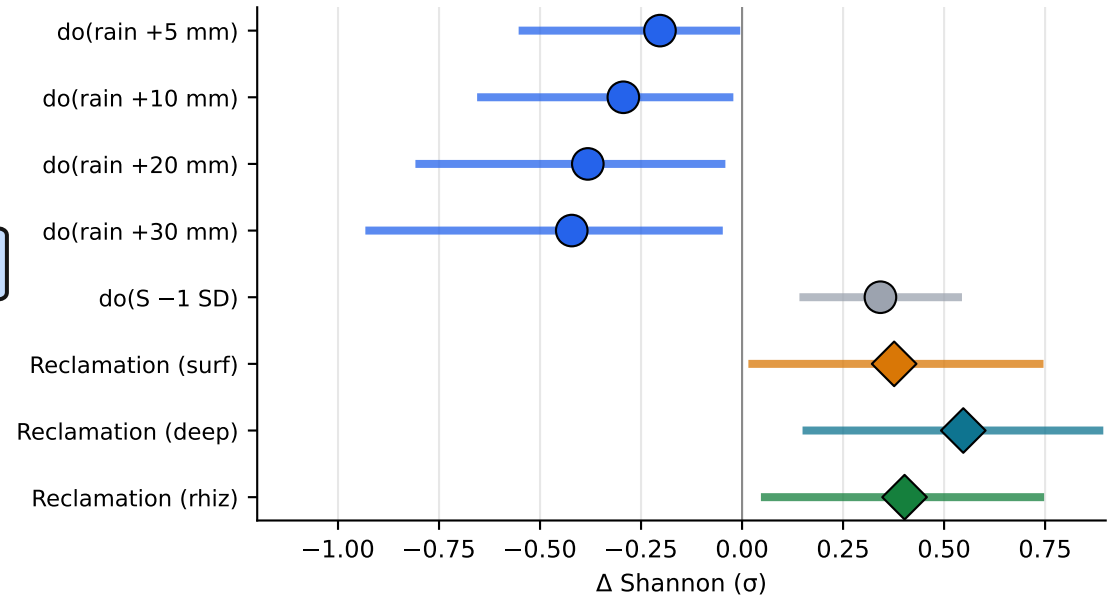


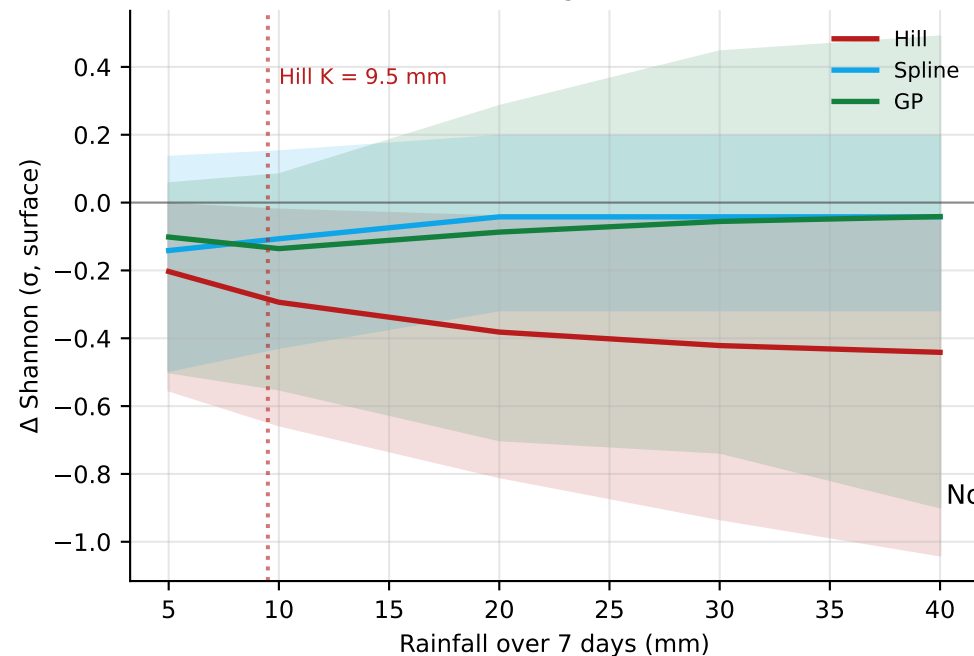
(a) Causal mediation: salinity $\rightarrow$ CSP1-2 $\rightarrow$ Shannon



(b) Interventional posteriors: rainfall, salinity, reclamation (Bayesian state-space digital twin, 95% CrI)



(c) Rainfall dose-response: three functional forms consistent at small mm, diverge at >20 mm



(d) JSDM knockout: top 12 genera affected by CSP1-2 removal (propagates to Δ Shannon ≈ -0.35 via biotic associations)

